

3D Game Assets

Computing and Mathematics

May, 2026

3D Game Assets (A11483)

Short Title:	3D Game Assets	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Game Development	
Author	PMCINERNEY	
Credits	5	Level: Introductory

Description of Module / Aims

This module serves as an introduction to 3D modelling for game asset designers and comprises an emphasis on the modelling pipeline, from concept to model creation to rigging and skinning, and techniques used for the exporting of 3D models to be used in a game engine.

Programmes

COMP -0645	BSc (Hons) in Applied Computing (WD_KACCM_B)	2 / 4 / E
COMP -0645	BSc (Hons) in Applied Computing (WD_KCOMP_B)	2 / 4 / E
COMP -0645	BSc (Hons) in Computer Science (WD_KCMSC_B)	2 / 4 / E

Indicative Content

- Components of an industry-standard 3D modelling platform
- Polygon, subdivision, and NURB modelling
- Rigging and skinning
- Texture unwrapping and baking
- Techniques for exporting models, textures and collisions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify and navigate the main components and features of an industry standard 3D modelling production platform.
2. Compare various modelling approaches, i.e, polygon, sub-division and NURB-based modelling.
3. Describe the fundamentals of the 3D modelling workflow, from concept to a rigged and skinned model.
4. Construct a textured and rigged game asset in a format ready to be used by a game engine.

Learning and Teaching Methods

- Interactive and open-forum lectures.
- Class discussions and presentations.
- Problem-based learning activities.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
<i>In-Class Assessment</i>	20%	1,2
<i>Presentation</i>	30%	3
<i>Project</i>	50%	3,4

Assessment Criteria

- <40%:** Inability to identify and navigate the main components of a 3D production software platform. Inability to critically evaluate techniques used in the creation and exporting of 3D computer graphics models.
- 40%-49%:** Ability to identify and navigate the main components of a 3D production software platform. Ability to describe key concepts of the 3D modelling pipeline and techniques used for exporting.
- 50%-59%:** Ability to discuss and critically evaluate key concepts of the 3D modelling pipeline and techniques used for exporting.
- 60%-69%:** Be able to solve problems within the framework of the 3D modelling pipeline by experimenting with appropriate skills and tools of the 3D production software.
- 70%-100%:** All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of problems through the use and modification of appropriate skills and tools of the 3D production software.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Supplementary Material

- Lanier, L. _Advanced Maya Texturing and Lighting_. 3rd ed. New York: Sybex, 2015.
- Palamar, T. _Mastering Autodesk Maya_. New York: Sybex, 2016.

Requested Resources

- Room Type: Computer Lab